

Web Development Assignment

1. Difference Between Frontend, Backend, and Full-Stack Development

- **Frontend Development:** Develops the user interface (UI) using HTML, CSS, and JavaScript. It is the part users interact with directly.
- **Backend Development:** Handles server-side logic, databases, authentication, and APIs using technologies such as Node.js, Java, Python, or PHP.
- **Full-Stack Development:** Combines both frontend and backend development to build complete web applications.
- **Real-world Example (Amazon):** Frontend displays product pages and search; Backend processes orders, payments, and user accounts; Full-stack developers work across both layers.

2. Client–Server Model

- Client (Browser) → HTTP Request → Web Server → HTTP Response (HTML, CSS, JavaScript) → Browser displays the webpage.

3. How a Browser Requests and Displays a Web Page

- User enters a website URL.
- Browser sends an HTTP/HTTPS request.
- DNS converts the domain name into an IP address.
- The request reaches the web server.
- The server processes the request and retrieves data.
- The server sends HTML, CSS, JavaScript, and images.
- The browser renders and displays the webpage.

4. Tools Required for Web Development Environment

- **Visual Studio Code:** Code editor.
- **Google Chrome:** Testing and debugging webpages.

- **Git:** Version control.
- **GitHub:** Online repository hosting.
- **Node.js:** JavaScript runtime.
- **Live Server:** Instantly previews changes.
- **Browser Developer Tools:** Debugging and inspecting web pages.

5. What is a Web Server?

- A web server stores website files and delivers them to users through HTTP/HTTPS.
- **Examples:** Apache HTTP Server, Nginx, Microsoft IIS, LiteSpeed, Node.js (Express).

6. Roles in a Project

- **Frontend Developer:** Builds the UI and improves user experience.
- **Backend Developer:** Develops server logic, APIs, and database connectivity.
- **Database Administrator (DBA):** Designs, secures, backs up, and optimizes databases.

7. VS Code Setup

- Install Visual Studio Code.
- Install extensions: Live Server, Prettier, HTML CSS Support, JavaScript (ES6) Snippets.
- Create **index.html**, **style.css**, and **script.js**.
- Attach a screenshot of your configured VS Code setup.

8. Static vs Dynamic Websites

- **Static Website:** Fixed content with no database. Example: Portfolio website.
- **Dynamic Website:** Content changes dynamically using a backend and database. Example: Facebook or Amazon.

9. Five Web Browsers and Rendering Engines

- Google Chrome — Blink
- Microsoft Edge — Blink
- Mozilla Firefox — Gecko
- Apple Safari — WebKit
- Opera — Blink
- **Rendering Engine:** Converts HTML, CSS, and JavaScript into the visual webpage displayed in the browser.

10. Basic Web Architecture Flow

- Client (Browser) → Web Server → APIs → Database → Web Server → Browser (Response Display).